

Adaptive Multi-Agent Artificial Intelligence for Real-Time Financial Market Decision-Making

Using Reinforcement Learning, Large Language Models and Regime Detection

PhD Research Proposal Outline

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Working document for supervisor discussion and proposal development

Proposal at a Glance

Working title	Adaptive Multi-Agent Artificial Intelligence for Real-Time Financial Market Decision-Making Using Reinforcement Learning, Large Language Models and Regime Detection
Academic alternative	Adaptive Multi-Agent Artificial Intelligence for Regime-Aware Financial Market Decision-Making under Non-Stationarity and Uncertainty
Candidate	David Maguire
Proposed award	Doctor of Philosophy
Proposed duration	Three years, subject to institutional requirements
Primary domain	Artificial intelligence, quantitative finance and financial-market decision systems
Core methods	Regime detection, financial NLP and LLMs, reinforcement learning, multi-agent systems, uncertainty estimation and risk control
Proposed institutions	[Insert university or department]
Potential supervisors	[Insert potential supervisors]

Executive Summary

This proposal investigates whether a coordinated system of specialised artificial-intelligence agents can make financial-market decisions that are more adaptive, robust and auditable than conventional static models. The system would combine probabilistic market-regime detection, large-language-model interpretation of financial narratives, reinforcement learning for sequential decisions and deterministic risk controls.

The research is motivated by the non-stationary nature of financial markets. Relationships between price, volatility, liquidity, macroeconomic conditions and investor behaviour change over time. A model developed in one environment can deteriorate when the market transitions to another. The proposed architecture therefore treats regime awareness, uncertainty and controlled adaptation as central design requirements rather than optional model features.

Central proposition

Financial-market decision systems should not rely on one static predictive model. They should operate as adaptive systems in which specialised agents identify market regimes, interpret new information, select actions, control risk and learn from outcomes.

Contents

Section	Focus
1. Introduction and background	Context, motivation and the non-stationary nature of financial markets.
2. Research problem	Limitations of static, fragmented and weakly adaptive approaches.
3. Research aim and objectives	Overall aim and seven specific objectives.
4. Research questions and hypotheses	Principal question, supporting questions and testable hypotheses.
5. Proposed multi-agent architecture	Roles, boundaries, interactions and decision workflow.
6. Theoretical and conceptual foundations	Finance, latent-state models, RL, multi-agent systems and LLMs.
7. Proposed methodology	Data, baselines, regime modelling, LLMs, RL and integration.
8. Evaluation framework	Predictive, financial, robustness and agent-specific measures.
9. Expected contribution and significance	Methodological, empirical, evaluation and governance contributions.
10. Proposed chapter structure	Indicative ten-chapter thesis structure.
11. Indicative three-year plan	Sequencing of foundations, components, integration and writing.
12. Scope, limitations and research risks	Scope controls, anticipated limitations and mitigations.
13. Indicative technical roadmap	Priority mathematical and machine-learning topics.
14. Concluding proposition	Recommended academic positioning of the research.
Appendix A	Key equations and formal definitions.

Document status

This is a structured proposal outline for supervisor discussion. Institution-specific requirements, the final literature base, data-access arrangements and formal ethics or governance sections should be completed during proposal development.

1. Introduction and Background

Financial markets are increasingly influenced by automated trading systems, quantitative strategies and artificial intelligence. These systems react rapidly to prices, order flow, volatility, economic announcements, financial news and the actions of other participants. At the same time, the market environment is neither stable nor fully observable. Volatility, liquidity, correlations, investor behaviour and the predictive value of signals change across time.

Traditional forecasting and trading models are often static. They are trained on historical relationships and generally assume that the underlying structure remains sufficiently stable. In practice, a model that performs well in a calm, trending market may deteriorate in a high-volatility or liquidity-stressed period. The challenge is therefore not solely to predict the next market movement, but to recognise when the decision environment has changed and adapt accordingly.

- Bullish and bearish market phases.
- High- and low-volatility periods.
- Trending and mean-reverting conditions.
- Liquidity crises and market dislocations.
- Economic cycles and changing monetary-policy environments.
- Geopolitical shocks and company-specific events.
- Periods dominated by systematic or machine-driven activity.

The proposed research would examine whether a modular multi-agent AI system can identify these changing conditions, interpret structured and unstructured information, adapt decisions through reinforcement learning and manage risk more effectively than conventional single-model approaches.

2. Research Problem

Existing financial-market decision models frequently exhibit five related weaknesses:

1. Static model behaviour: relationships learned in one period may continue to be applied after the environment has changed.
2. Limited regime awareness: models may identify price patterns without explicitly estimating whether the market is trending, mean-reverting, volatile, illiquid or undergoing structural change.
3. Fragmented information processing: prices, macroeconomic data, financial news, corporate disclosures and geopolitical developments are often analysed separately.
4. Weak adaptation mechanisms: supervised models do not directly learn which sequences of decisions improve longer-term risk-adjusted outcomes.
5. Limited coordination between specialised functions: forecasting, news interpretation, position sizing, risk control and execution are frequently concentrated in one model or loosely connected pipeline.

Central research problem

Can a coordinated system of specialised AI agents adapt to changing market regimes and make more robust financial-market decisions than static single-model approaches?

3. Research Aim and Objectives

3.1 Overall research aim

To design, develop and evaluate an adaptive multi-agent artificial-intelligence framework that combines financial-market regime detection, large-language-model interpretation, reinforcement learning and formal risk controls to support real-time financial-market decision-making.

3.2 Research objectives

Objective	Description
Objective 1	Develop a market-regime detection framework capable of identifying latent and observable market states using statistical and machine-learning methods.
Objective 2	Investigate whether market microstructure and high-frequency features can identify periods characterised by elevated systematic or algorithmic activity.
Objective 3	Develop an LLM-based news and narrative agent that converts unstructured information into structured, timestamp-safe market features.
Objective 4	Develop a reinforcement-learning agent that adapts exposure, position size, timing and holding decisions from risk-adjusted outcomes.
Objective 5	Design a multi-agent coordination framework governing information access, communication, confidence, conflict resolution and credit assignment.
Objective 6	Incorporate deterministic risk management capable of scaling, rejecting or overriding proposed actions.
Objective 7	Evaluate the complete architecture against transparent, econometric, supervised-learning, single-agent and regime-unaware benchmarks.

4. Research Questions and Hypotheses

4.1 Principal research question

Principal research question

Can an adaptive multi-agent artificial-intelligence system combining probabilistic regime detection, large-language-model interpretation and risk-constrained reinforcement learning improve the robustness and adaptability of financial-market decisions under non-stationary conditions?

4.2 Supporting research questions

Ref.	Research question
RQ1	Which statistical and machine-learning methods most reliably identify financial-market regimes and structural changes out of sample?
RQ2	Does conditioning forecasting and decision models on estimated regime probabilities improve predictive calibration and economic performance?
RQ3	Can LLMs extract structured, time-sensitive information from financial news, economic announcements and geopolitical events that adds value beyond price, volume and volatility data?

Ref.	Research question
RQ4	Can risk-constrained reinforcement learning improve position sizing, exposure management or execution compared with transparent deterministic policies?
RQ5	Does a modular multi-agent architecture outperform equivalent single-model and single-agent systems after transaction costs and risk are considered?
RQ6	Can market microstructure and high-frequency features identify periods characterised by elevated algorithmic or machine-driven activity?
RQ7	How stable are the system's decisions across different assets, market regimes, periods and repeated experimental trials?
RQ8	Can individual agent contributions and final decisions be made sufficiently explainable, reproducible and auditable for financial-market applications?

4.3 Proposed hypotheses

Ref.	Hypothesis
H1	Regime-conditioned forecasting models will demonstrate greater out-of-sample stability than equivalent regime-unaware models.
H2	Structured information extracted from financial text using an LLM will provide incremental predictive or risk-management value beyond price, volume and volatility features.
H3	Risk-constrained reinforcement learning will improve dynamic exposure management compared with static position-sizing rules.
H4	A coordinated multi-agent system will provide better risk-adjusted performance than an equivalent single-agent architecture.
H5	The greatest benefit of the multi-agent system will arise during regime transitions and periods of elevated uncertainty rather than stable conditions.
H6	Hard-coded risk constraints will improve stability and practical viability, even where they reduce gross returns.

5. Proposed Multi-Agent Architecture

The architecture is modular. Each agent has a bounded function, defined data access and an auditable output. The LLM would not be permitted to place trades directly. Final actions would require risk approval and would be evaluated against deterministic baselines.

No.	Agent	Primary role	Illustrative methods and outputs
1	Market Regime Agent	Identifies the current market state and estimates regime probabilities.	HMMs, Markov-switching models, Bayesian change-point detection, clustering, GARCH states, sequence models.
2	AI and Market Behaviour Detection Agent	Detects unusual behaviour consistent with systematic or algorithmic activity.	Order-book imbalance, trade intensity, tick volume, spread changes, volatility bursts, price acceleration and liquidity withdrawal.
3	News and Narrative Agent	Transforms unstructured information into bounded, auditable market features.	Event class, direction, sentiment, novelty, urgency, affected assets, time horizon and source reliability.
4	Reinforcement-Learning Trading Agent	Selects exposure, timing and position-management actions from the combined state.	Discrete or continuous actions, risk-sensitive rewards, PPO, actor-critic and constrained RL.
5	Risk Management Agent	Approves, scales or rejects proposed actions.	Exposure, leverage, volatility, drawdown, liquidity, concentration, Expected Shortfall and transaction-cost controls.
6	Review and Learning Agent (optional)	Attributes outcomes, tests compliance and informs future learning.	Agent contribution, regime accuracy, misleading signals, post-trade review and decision audit trail.

5.1 Conceptual decision process



5.2 Illustrative agent outputs

Regime Agent	Probabilities of trend, mean reversion, high volatility, liquidity stress and structural break.
Behaviour Agent	Indicators of unusual trade intensity, spread widening, liquidity withdrawal or cross-market reaction speed.
News Agent	Event class, affected assets, directional implication, confidence, novelty, expected duration and source reliability.
RL Agent	Proposed exposure, position size, entry or exit, holding period and action confidence.
Risk Agent	Approve, reject or scale; reason code; applicable limit; resulting exposure.
Review Agent	Attribution of profit or loss, agent contribution, regime accuracy and compliance assessment.

6. Theoretical and Conceptual Foundations

6.1 Financial-market theory

- Market efficiency and limits to efficiency.
- Behavioural finance and bounded rationality.
- Adaptive Markets Hypothesis.
- Market microstructure, liquidity and price discovery.
- Volatility clustering and regime switching.
- Systemic risk and contagion.

6.2 Statistical learning and latent-state modelling

- Time-series forecasting and non-stationarity.
- Hidden Markov and Markov-switching models.
- Bayesian inference and change-point detection.
- State-space models and filtering.
- Probability calibration and uncertainty estimation.
- Ensemble learning and model combination.

6.3 Reinforcement learning

The financial decision process may be represented as a Markov Decision Process:

$$M = (S, A, P, R, \gamma)$$

Where S represents states, A actions, P the transition process, R rewards and γ the discount factor. Because the true market regime is not directly observable, the final framework may be better represented as a Partially Observable Markov Decision Process in which the agent maintains a belief over latent states.

$$b_t(s) = P(S_t = s \mid o_{1:t}, a_{1:t-1})$$

- Q-learning and Deep Q-Networks.
- Policy-gradient methods.
- Actor-critic algorithms.
- Proximal Policy Optimisation.
- Offline reinforcement learning.
- Constrained and risk-sensitive reinforcement learning.

6.4 Multi-agent systems

- Cooperative agents and Markov games.
- Centralised training with decentralised execution.
- Shared and individual reward functions.
- Communication protocols and information permissions.
- Credit assignment and counterfactual contribution.
- Coordination, redundancy and conflict resolution.

6.5 Large language models

- Financial sentiment and event extraction.
- Retrieval-augmented generation.
- Tool-using and constrained agents.
- Temporal information integrity.
- Hallucination and consistency controls.
- Traceable evidence and model governance.

7. Proposed Methodology

7.1 Research design

The project would use a staged, empirical design. Each component would first be evaluated independently, then integrated incrementally. This approach permits attribution of performance and avoids attributing gains to the full system when only one component is responsible.

7.2 Phase 1: Literature review

- Financial-market regime detection and structural breaks.

- Algorithmic trading and market microstructure.
- Reinforcement learning in finance.
- Multi-agent systems and agent coordination.
- LLMs and financial natural-language processing.
- Explainable AI, model risk and governance.
- Backtest overfitting and statistical validation.

7.3 Phase 2: Data collection and preparation

Dataset	Illustrative content
Market data	Open, high, low, close, returns, volume, realised volatility, spreads, order-flow and tick-level information where available.
Fundamental data	Earnings, valuation measures, balance-sheet indicators, earnings surprises and analyst expectations.
Macroeconomic data	Inflation, interest rates, unemployment, GDP, bond yields, credit spreads and policy decisions.
Textual data	Financial news, company announcements, earnings calls, regulatory filings, central-bank speeches and geopolitical news.
Cross-asset data	Equity indices, volatility indices, bonds, currencies, commodities and sector indices.
Temporal controls	Every observation requires an available-at timestamp to prevent look-ahead bias.

7.4 Phase 3: Baseline models

- Buy-and-hold and transparent rule-based strategies.
- Linear and logistic regression.
- ARIMA and GARCH.
- Random Forest, XGBoost or LightGBM.
- LSTM or other sequence models.
- Conventional regime-switching models.

7.5 Phase 4: Regime modelling

- Rule-based volatility states.
- K-means clustering and Gaussian mixture models.
- Hidden Markov Models.
- Markov-switching models.
- Bayesian online change-point detection.
- Deep latent-state models.

Regime models would be evaluated for stability, persistence, economic interpretation, detection delay, out-of-sample consistency and downstream usefulness.

7.6 Phase 5: LLM and narrative modelling

The LLM would convert text into bounded structured outputs rather than free-form trading instructions. Outputs would include event category, directional implication, confidence, novelty, expected duration, affected instruments, source reliability and supporting evidence.

7.7 Phase 6: Reinforcement-learning development

The state representation could combine recent returns, volatility, volume, technical indicators, regime probabilities, news features, macroeconomic variables, current holdings, profit and loss, drawdown, liquidity and spread.

Actions may be discrete:

$$A_t \in \{-1, 0, +1\}$$

or continuous:

$$A_t \in [-1, 1]$$

The reward would penalise risk, drawdown, transaction costs and turnover rather than maximise raw profit alone:

$$R_t = r_t - \lambda_1 \sigma_t^2 - \lambda_2 DD_t - \lambda_3 TC_t - \lambda_4 TO_t$$

Candidate algorithms include Q-learning, Deep Q-Networks, actor-critic methods, PPO, Soft Actor-Critic and constrained reinforcement learning.

7.8 Phase 7: Multi-agent integration

6. Regime Agent only.
7. Regime plus supervised prediction.
8. Regime plus LLM News Agent.
9. Regime, news and RL Agent.
10. Full system with Risk Agent.
11. Full system with Review Agent.

7.9 Phase 8: Backtesting and out-of-sample evaluation

- Chronological train, validation and test periods.
- Walk-forward and rolling-window evaluation.
- Purged cross-validation and embargo periods.
- Transaction costs, spreads, slippage and delayed execution.
- Regime-specific evaluation and stress tests.
- Parameter sensitivity and repeated random seeds.
- Testing across distinct market and monetary-policy environments.

8. Evaluation Framework

8.1 Predictive metrics

- Accuracy, precision, recall and F1 score.
- ROC-AUC and precision-recall AUC.
- Log loss and Brier score.
- Probability calibration and expected calibration error.
- Quantile loss and prediction-interval coverage.

8.2 Financial metrics

- Annualised and cumulative return.
- Volatility, Sharpe, Sortino and Calmar ratios.

- Maximum drawdown and Expected Shortfall.
- Turnover, transaction costs and implementation shortfall.
- Profit factor, hit rate and exposure utilisation.

8.3 Robustness and statistical validation

- Performance by regime and asset.
- Sensitivity to costs, reward design and model parameters.
- Block bootstrap confidence intervals.
- Multiple-testing corrections.
- Probability of Backtest Overfitting.
- Deflated Sharpe Ratio.
- Reality-check or superior-predictive-ability tests.

8.4 Agent-specific evaluation

- Incremental contribution of each agent.
- Decision consistency under repeated trials.
- Agent disagreement and conflict-resolution outcomes.
- Communication efficiency and latency.
- Frequency and value of risk overrides.
- Attribution of gains and losses through ablation tests.

9. Expected Contribution and Significance

9.1 Methodological contribution

A unified architecture combining regime detection, LLM-derived information, reinforcement learning, multi-agent coordination and formal risk management.

9.2 Empirical contribution

Evidence on whether each component provides incremental value after transaction costs, model-selection bias, regime change and realistic execution assumptions are considered.

9.3 Evaluation contribution

A structured framework for evaluating multi-agent financial systems using ablation testing, agent attribution, decision consistency, temporal integrity, risk overrides and regime-specific performance.

9.4 Practical and governance contribution

A controlled approach to agentic AI in finance that retains deterministic limits, human oversight, traceable decisions, auditable data and explicit model-risk management.

10. Proposed Chapter Structure

Chapter	Title	Purpose
Chapter 1	Introduction	Research context, problem, aim, questions and expected contribution.
Chapter 2	Literature Review	Regime detection, RL, LLMs, multi-agent systems, market microstructure and AI governance.

Chapter	Title	Purpose
Chapter 3	Research Design and Data	Datasets, feature engineering, timestamp controls and experimental design.
Chapter 4	Financial-Market Regime Detection	Development and comparison of regime-detection approaches.
Chapter 5	LLM-Based Narrative and Event Analysis	Design and evaluation of the news and narrative agent.
Chapter 6	Risk-Constrained Reinforcement Learning	RL environment, reward functions and adaptive decision policies.
Chapter 7	Multi-Agent Architecture	Communication, coordination, conflict resolution and risk controls.
Chapter 8	Empirical Evaluation	Out-of-sample results, ablations, regime analysis and robustness.
Chapter 9	Explainability, Governance and Implications	Auditability, reproducibility, model risk and practical application.
Chapter 10	Conclusions	Findings, contributions, limitations and future research.

11. Indicative Three-Year Research Plan

Period	Indicative activity
Year 1: Foundations and regime modelling	Complete the literature review; finalise questions; obtain and prepare datasets; establish baselines; develop regime models; prepare the first study or paper.
Year 2: LLM and reinforcement-learning components	Develop and evaluate the News Agent; construct the RL environment; test reward functions and risk constraints; prepare a second study or paper.
Year 3: Multi-agent integration and thesis completion	Integrate agents; conduct ablation and out-of-sample testing; complete robustness, governance and explainability analysis; write and submit the thesis.

12. Scope, Limitations and Research Risks

12.1 Scope controls

- Begin with one or a small number of liquid markets before attempting broad cross-asset generalisation.
- Use the LLM as an information-processing agent, not an unconstrained execution authority.
- Evaluate position sizing or exposure management before attempting a fully autonomous trading system.
- Prioritise robustness, attribution and governance over maximum backtest return.

12.2 Anticipated limitations

- Limited access to institutional order-book and news data.
- Difficulty proving that observed behaviour is specifically caused by AI rather than broader systematic trading.
- Non-stationarity and limited examples of extreme regimes.
- High sensitivity of RL models to reward design and training configuration.
- LLM model drift, vendor changes and reproducibility constraints.
- Computational cost and the risk of over-complexity.

12.3 Principal research risks and mitigations

Risk	Mitigation
Look-ahead bias	Enforce available-at timestamps, delayed execution and frozen historical information sets.
Backtest overfitting	Use simple baselines, purged walk-forward testing, multiple-testing controls, PBO and Deflated Sharpe analysis.
Agent complexity without value	Integrate incrementally and require ablation evidence for every component.
Unstable RL behaviour	Use constrained action spaces, hard risk limits, offline evaluation and conservative reward design.
LLM hallucination	Require schema-constrained outputs, evidence retrieval, source attribution and deterministic validation.
Weak explainability	Log inputs, outputs, confidence, risk overrides and final decision reasons for each agent.

13. Indicative Technical Roadmap

12. Markov chains and transition probabilities.
13. Hidden Markov Models and regime-duration analysis.
14. Bayesian online change-point detection.
15. Markov Decision Processes and Bellman equations.
16. Q-learning, policy gradients and actor-critic methods.
17. Risk-sensitive reward design and constrained RL.
18. Probability calibration and conformal prediction.
19. Purged walk-forward validation, PBO and Deflated Sharpe Ratio.
20. Attention, transformers and structured financial NLP.
21. Multi-agent Markov games and credit assignment.
22. Market microstructure, transaction costs and execution modelling.
23. Full modular integration with ablation and governance testing.

14. Concluding Proposition

The proposal's central academic argument is that financial-market decision systems should be designed as adaptive, modular and risk-governed systems rather than as isolated predictive models. The contribution would not rest solely on producing a profitable strategy. It would assess whether regime awareness, structured textual information, sequential learning and specialised agent coordination improve robustness, explainability and decision quality under changing conditions.

Regime detection → Information interpretation → Adaptive decision-making → Risk control → Continuous evaluation

Recommended research positioning

Frame the project as a study of adaptive, regime-aware and governable financial decision systems. Avoid presenting the principal objective as the creation of a fully autonomous profitable trading system. The research value lies in controlled integration, incremental evidence and rigorous evaluation.

Appendix A: Key Equations and Formal Definitions

A.1 Markov Decision Process

$$M = (S, A, P, R, \gamma)$$

A formal representation of the state space, action space, transition process, reward function and discount factor.

A.2 Belief state under partial observability

$$b_t(s) = P(S_t = s \mid o_{1:t}, a_{1:t-1})$$

Represents the probability of the hidden market state given observations and previous actions.

A.3 Hidden-state transition probability

$$P_{ij} = P(S_{t+1} = j \mid S_t = i)$$

A.4 Expected regime duration

$$E[D_i] = 1 / (1 - p_{ii})$$

A.5 Discounted return

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

A.6 Q-learning update

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$$

A.7 Advantage function

$$A^{pi}(s, a) = Q^{pi}(s, a) - V^{pi}(s)$$

A.8 Risk-sensitive reward

$$R_t = r_t - \lambda_1 \sigma_t^2 - \lambda_2 DD_t - \lambda_3 TC_t - \lambda_4 TO_t$$

A.9 Team reward

$$R_t^{team} = \sum_{i=1}^N w_i R_{i,t}$$

A.10 Brier score

$$BS = (1/N) \sum_{i=1}^N (p_i - y_i)^2$$